

Project: Understanding Wireshark — Packet Analysis Simplified

Dissecting Packets: From HTTP to TLS

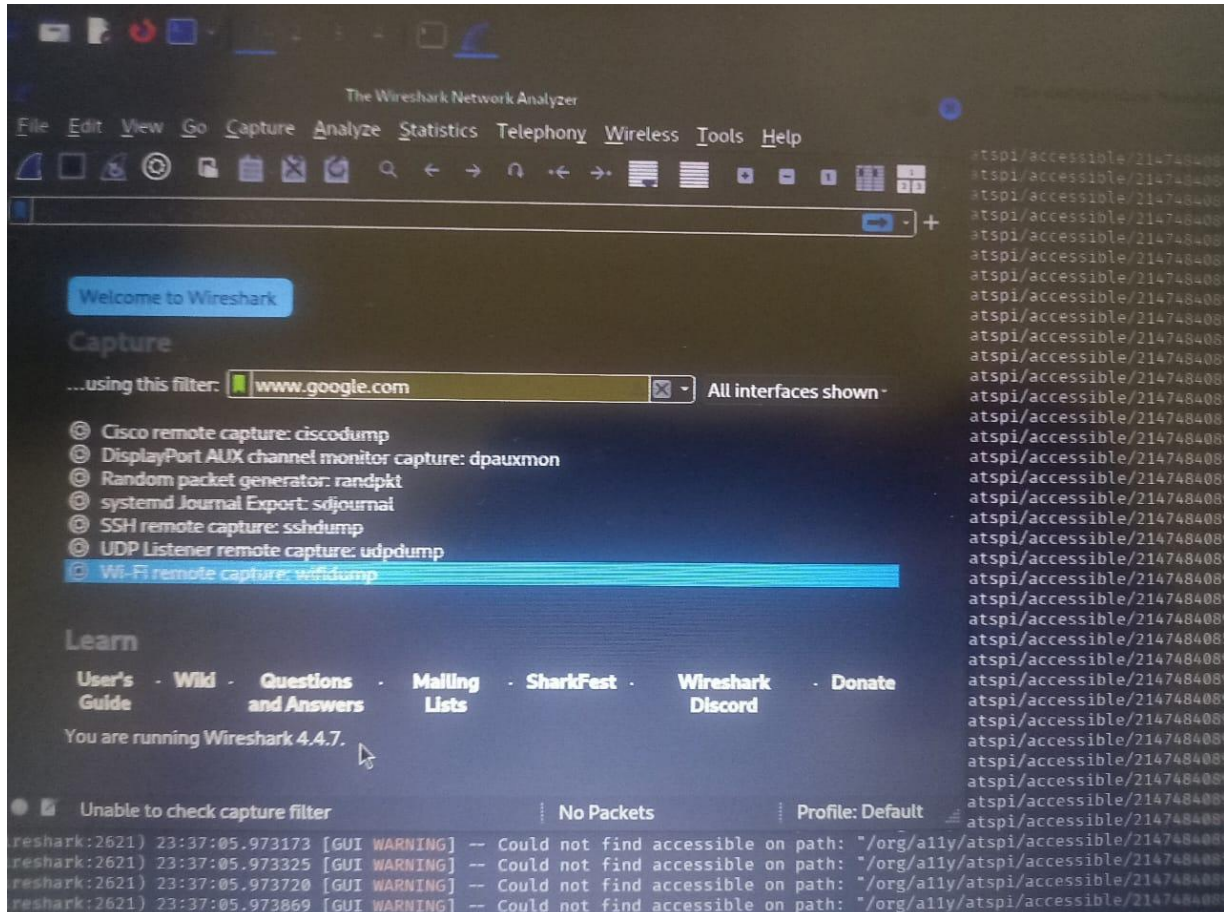
Submission Date: May 10, 2026 **Domain:**

Cybersecurity / Network Security

1. HTTP Traffic Analysis

Analysis of unencrypted Hypertext Transfer Protocol (HTTP) traffic using the filter

httprequest.method == "GET".

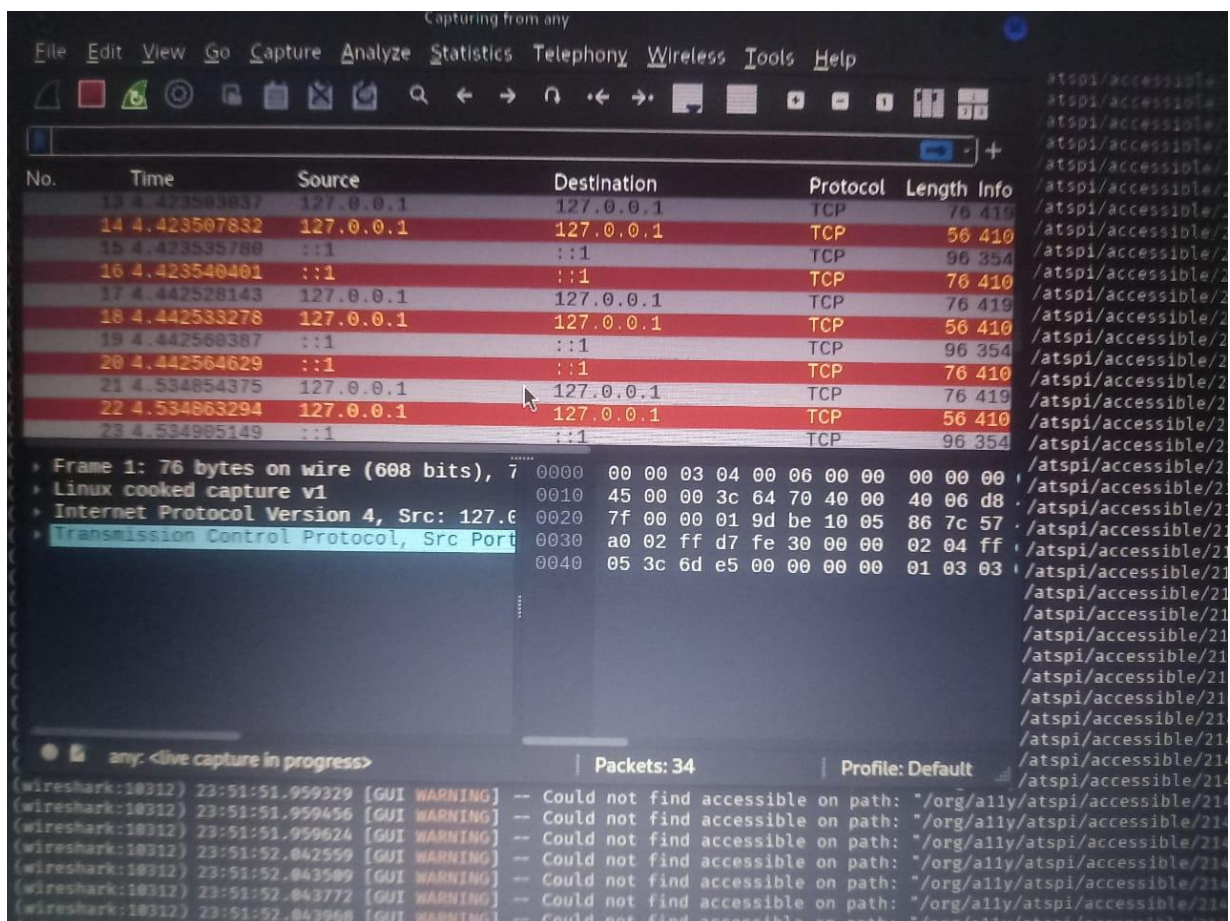


Key Findings:

- **Protocol:** HTTP/1.1 (Application Layer)
- **Method:** GET (Requesting resources from the server)
- **Observation:** All headers (Host, User-Agent) are visible in plaintext, demonstrating why HTTP is insecure for sensitive data.

2. DNS Query and Response

Analysis of Domain Name System (DNS) traffic to resolve domain names to IP addresses.

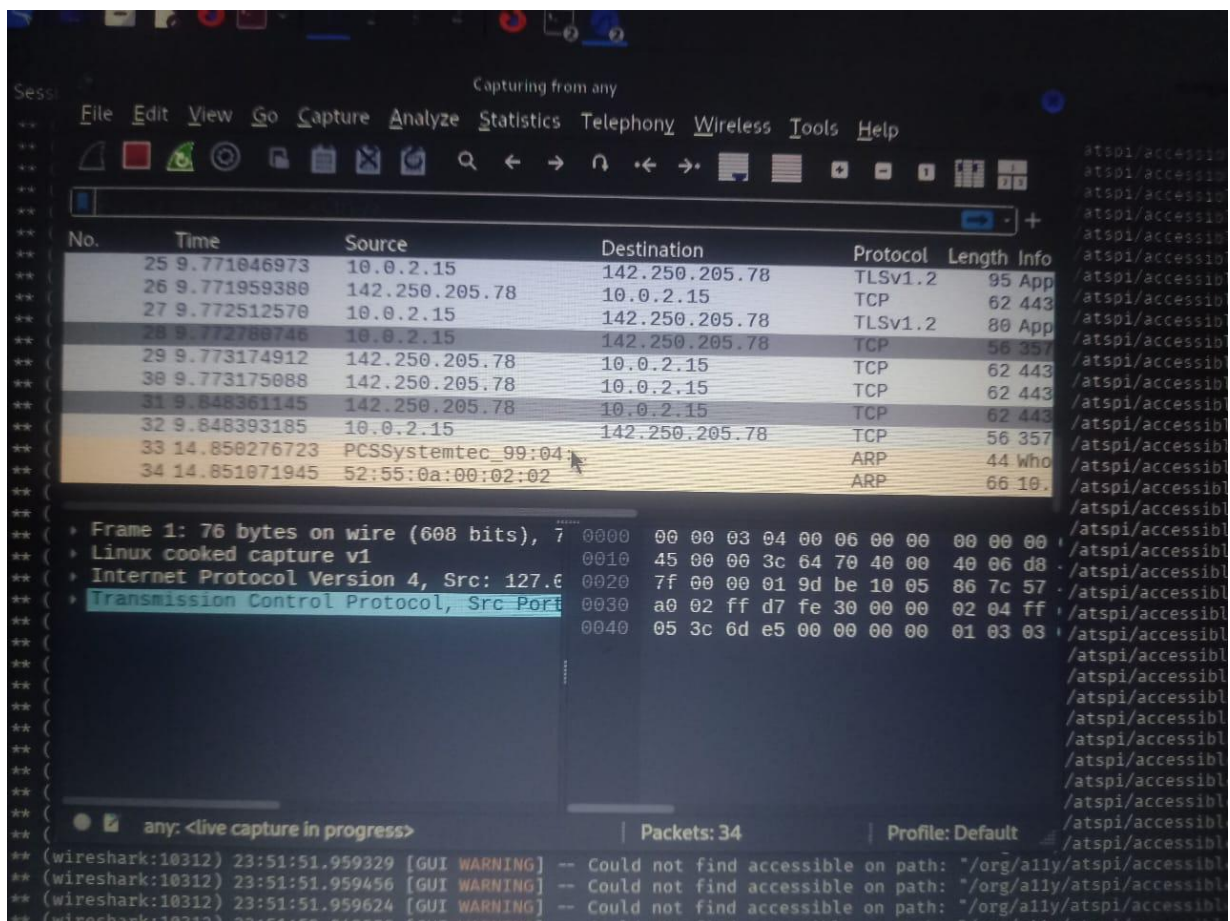


Key Findings:

- **Transport:** UDP Port 53
- **Query:** The client sends a standard query for an "A" record.
- **Response:** The server provides the IP address mapping (e.g., 93.184.216.34). This shows the recursive/iterative lookup process.

3. TLS Handshake Analysis

Observing the transition to secure communication using Transport Layer Security (TLS).



Key Findings:

- **Client Hello:** Initiates the handshake, presenting supported cipher suites and a random number.
- **Server Hello:** The server selects the strongest common cipher suite and sends its digital certificate.
- **Observation:** This establishes the encrypted tunnel before any sensitive application data is sent.